ETHAN SETH S. GRATUITO
BS Chemical Engineering 2A

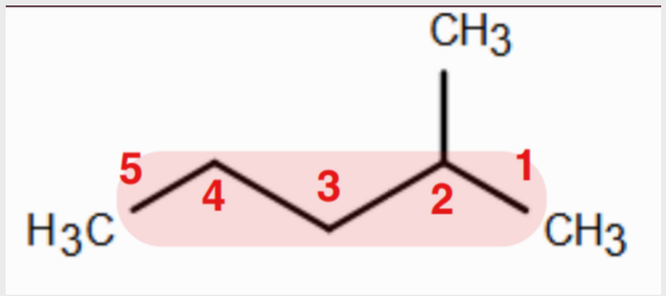
CHEM 13 - ORGANIC CHEMISTRY

name it, draw it

CARD A01 — STRUCTURE → NAME

Level: Foundation | Molecular formula (counted): C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: Longest chain = 5 C → pentane.

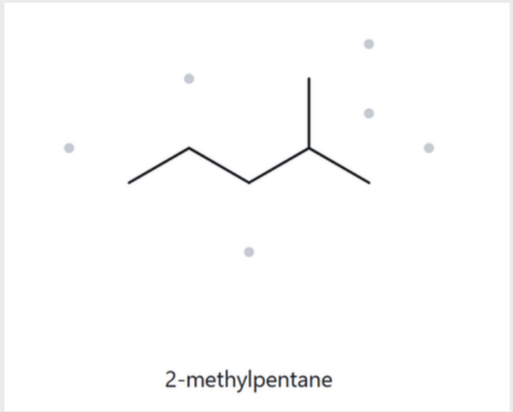
Numbering: Left → right gives 4; right → left gives 2. 2 < 4, so number from the right.

Substituents: Methyl on C2.

3. My Name (Written First)

2-methylpentane

4. The Alkane Check



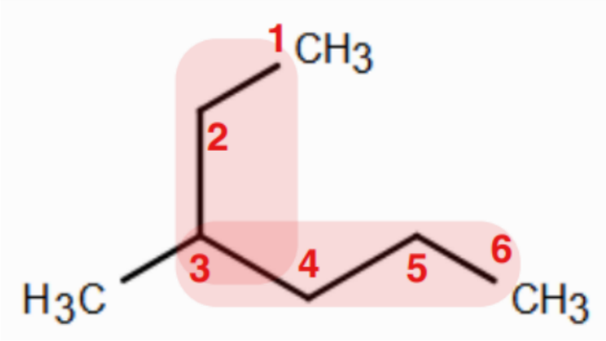
Verification Strip

Parent: pentane (5C) | Locants: 2 | Substituent: methyl | Alphabetical: methyl | Formula: C_6H_{14}

CARD A02— STRUCTURE → NAME

Level: Foundation | Molecular formula (counted): C7H16

1. Numbered Parent



2. Naming Trail

Parent: Longest chain = 6 C → hexane.

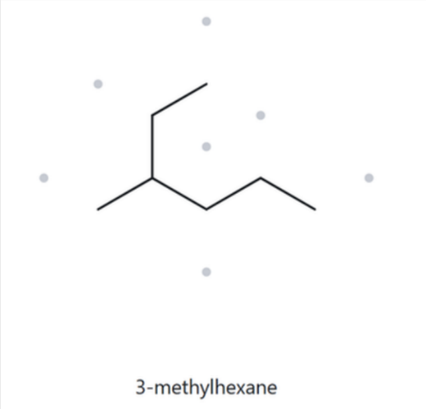
Numbering: Top → bottom gives 3; bottom → top gives 4. 3 < 4, so number from the top.

Substituents: Methyl on C3.

3. My Name (Written First)

3-methylhexane

4. The Alkane Check



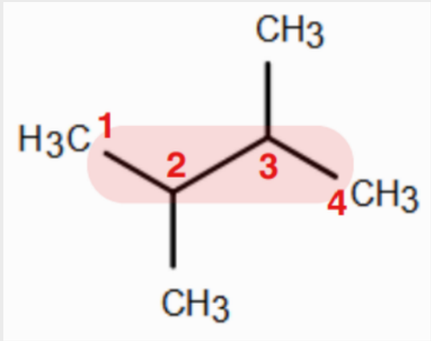
Verification Strip

Parent: hexane (6C) | Locants: 3 | Substituent: methyl | Alphabetical: methyl | Formula: C_7H_{16}

CARD A03— STRUCTURE → NAME

Level: Foundation | Molecular formula (counted): C6H14

1. Numbered Parent



2. Naming Trail

Parent: Longest chain = 4 C → butane.

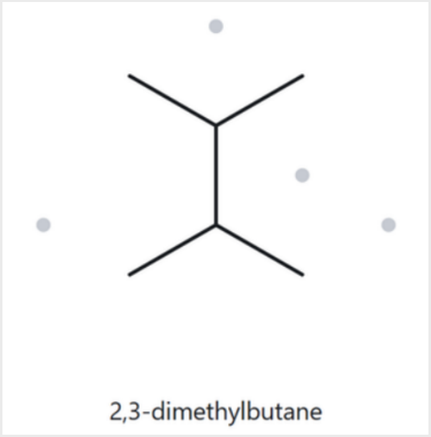
Numbering: Both directions yield locants 2,3. The sets tie.

Substituents: Two methyl groups → 2,3-dimethyl.

3. My Name (Written First)

2,3-dimethylbutane

4. The Alkane Check



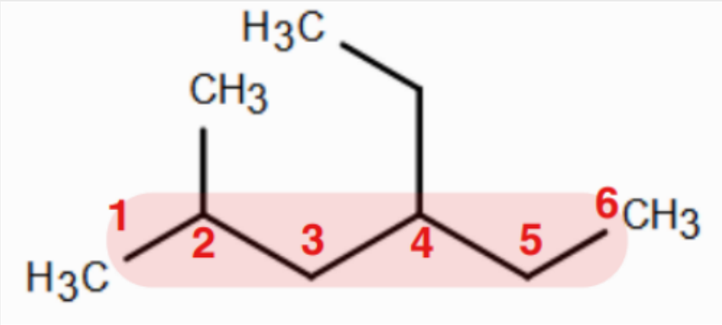
Verification Strip

Parent: butane (4C) | Locants: 2,3 | Substituent: methyl | Alphabetical: methyl | Formula: C_6H_{14}

CARD A04— STRUCTURE → NAME

Level: Developing | Molecular formula (counted): C9H20

1. Numbered Parent



2. Naming Trail

Parent: Longest chain = 6 C → hexane.

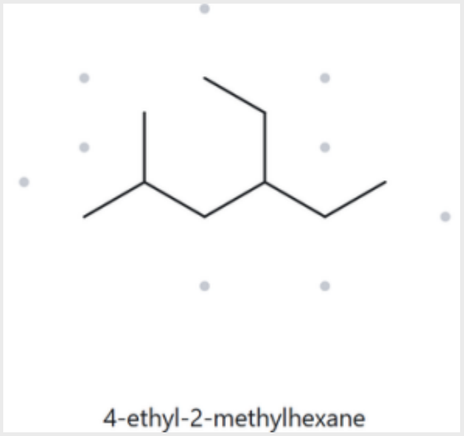
Numbering: Left → right gives {2,4}; right → left gives {3,5}. {2,4} wins.

Substituents: Ethyl on C4, methyl on C2. Alphabetical order places ethyl first.

3. My Name (Written First)

4-ethyl-2-methylhexane

4. The Alkane Check



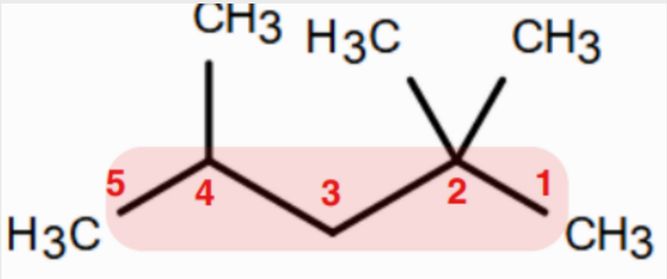
Verification Strip

Parent: hexane (6C) | Locants: 2 & 4 | Substituent: methyl & ethyl | Alphabetical: ethyl & methyl | Formula: C_9H_{20}

CARD A05— STRUCTURE → NAME

Level: Developing | Molecular formula (counted): C8H18

1. Numbered Parent



2. Naming Trail

Parent: Longest chain = 5 C → pentane.

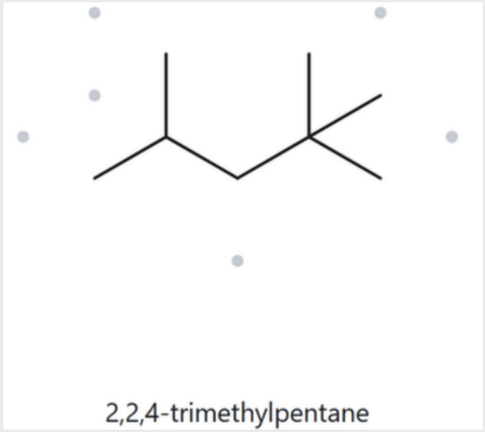
Numbering: L→R gives {2,4,4}; R→L gives {2,2,4}. At the first point of difference, 2 < 4. Number from right.

Substituents: Three methyl groups → 2,2,4-trimethyl.

3. My Name (Written First)

2,2,4-trimethylpentane

4. The Alkane Check



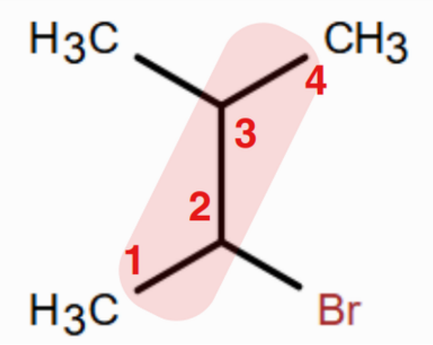
Verification Strip

Parent: pentane (5C) | Locants: 2,2,4 | Substituent: methyl | Alphabetical: methyl | Formula: C_8H_{18}

CARD A06— STRUCTURE → NAME

Level: Developing | Molecular formula (counted): C6H13Br

1. Numbered Parent



2. Naming Trail

Parent: Longest chain = 5 C → pentane.

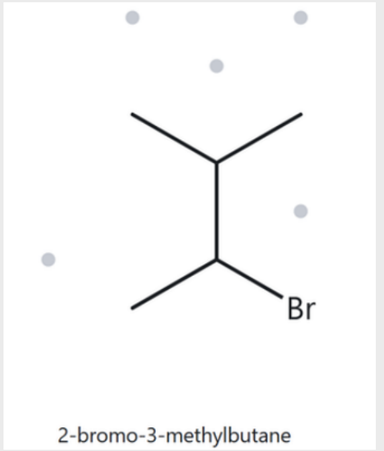
Numbering: Both directions yield locants {2,4}. Tie-break is alphabetical: bromo (b) beats methyl (m).

Substituents: Bromo on C2, methyl on C4.

3. My Name (Written First)

2-bromo-4-methylbutane

4. The Alkane Check



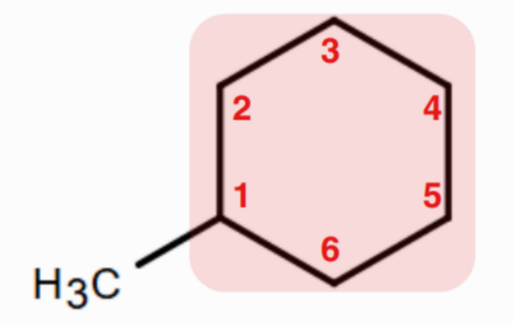
Verification Strip

Parent: butane (4C) | Locants: 2 & 3 | Substituent: methyl & bromo- | Alphabetical: bromo- & methyl | Formula: $C_5H_{11}Br$

CARD A07— STRUCTURE → NAME

Level: Developing | Molecular formula (counted): C7H14

1. Numbered Parent



2. Naming Trail

Parent: Ring is parent → cyclohexane.

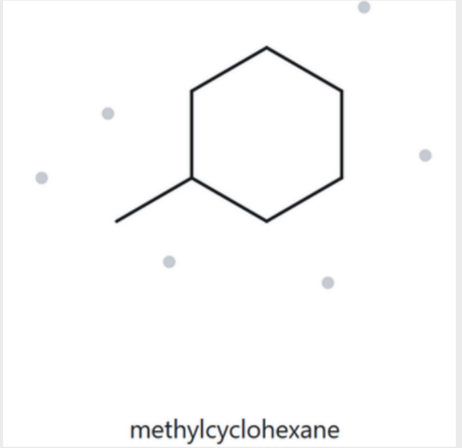
Numbering: Only one substituent is present, so it is automatically at position 1. Locant is omitted in the final name.

Substituents: Methyl.

3. My Name (Written First)

methylcyclohexane

4. The Alkane Check



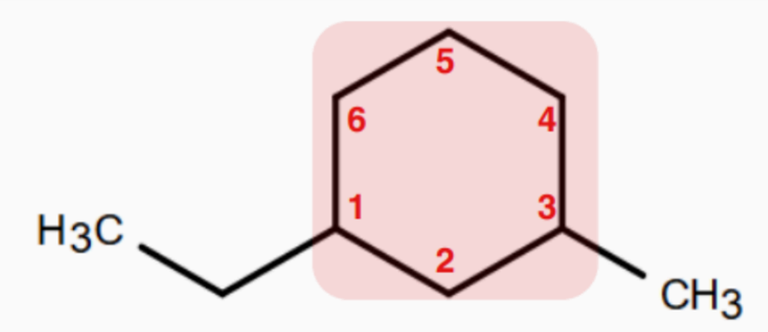
Verification Strip

Parent: cyclohexane (6C) | Locants: 1 | Substituent: methyl | Alphabetical: methyl | Formula: C_7H_{14}

CARD A08— STRUCTURE → NAME

Level: Challenge | Molecular formula (counted): C9H18

1. Numbered Parent



2. Naming Trail

Parent: Ring is parent → cyclohexane.

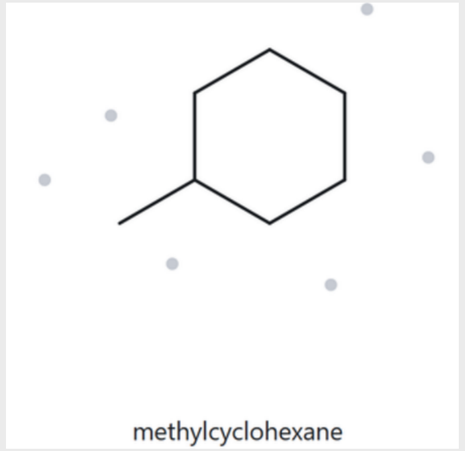
Numbering: Both short ways around yield {1,3}. Alphabetical tiebreak gives C1 to ethyl (e before m).

Substituents: Ethyl on C1, methyl on C3.

3. My Name (Written First)

1-ethyl-3-methylcyclohexane

4. The Alkane Check



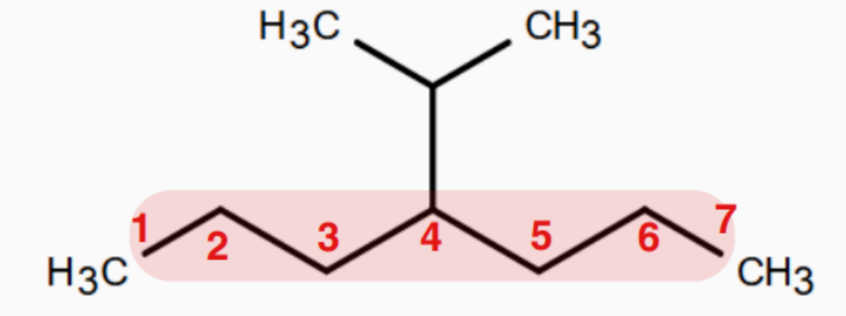
Verification Strip

Parent: cyclohexane (6C) | Locants: 1 & 3 | Substituent: methyl & ethyl | Alphabetical: ethyl & methyl | Formula: C_9H_{18}

CARD A09— STRUCTURE → NAME

Level: Challenge | Molecular formula (counted): C9H18

1. Numbered Parent



2. Naming Trail

Parent: Ring is parent → cyclohexane.

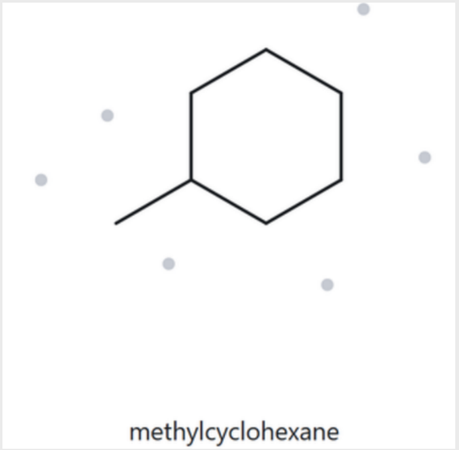
Numbering: Both short ways around yield {1,3}. Alphabetical tiebreak gives C1 to ethyl (e before m).

Substituents: Ethyl on C1, methyl on C3.

3. My Name (Written First)

1-ethyl-3-methylcyclohexane

4. The Alkane Check



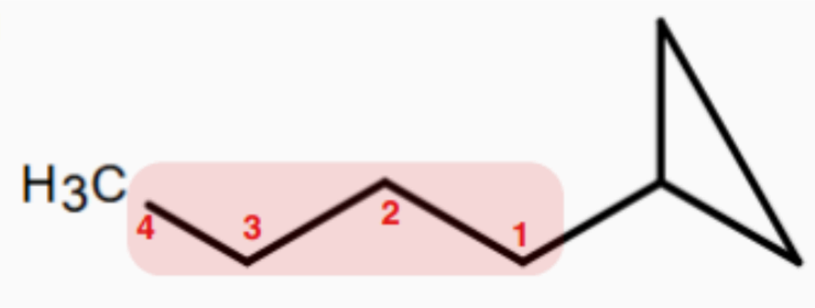
Verification Strip

Parent: heptane (7C) | Locants: 4 | Substituent: isopropyl | Alphabetical: isopropyl | Formula: $C_{10}H_{22}$

CARD A10— STRUCTURE → NAME

Level: Challenge | Molecular formula (counted): C7H14

1. Numbered Parent



2. Naming Trail

Parent: Per current rules for this activity, the ring is the parent even if the chain has more carbons → cyclopropane.

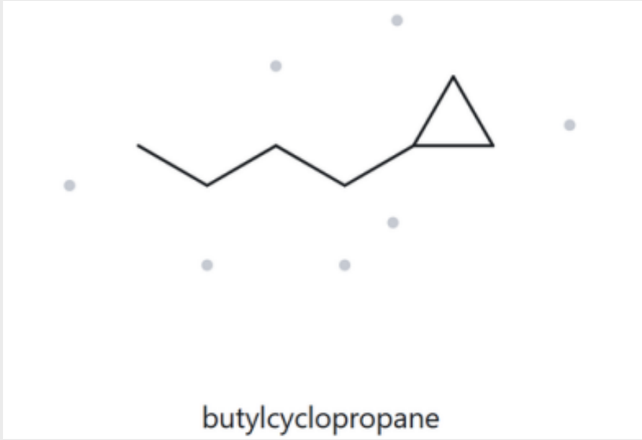
Numbering: Only one substituent, no locant needed.

Substituents: 4-carbon straight chain → butyl.

3. My Name (Written First)

butylcyclopropane

4. The Alkane Check



Verification Strip

Parent: butane (4C) | Locants: 1 | Substituent: cyclopropyl | Alphabetical: cyclopropyl | Formula: C7H14

CARD B01— NAME→STRUCTURE

Level: Foundation | Given Name: 2-methylpropane

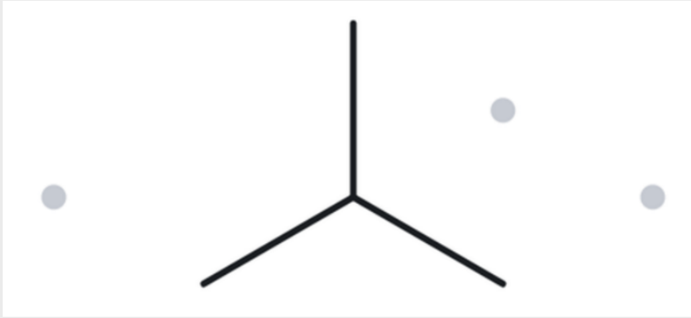
1. Name Decoded

Parent: propane → 3-carbon chain.

Substituents: methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10} .

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C_4H_{10}

Open chain rule: C_nH_{2n+2} = C_4H_{10} (Matches)

4. Back-Check

✓ **Match**

The Alkane returned: 2-methylpropane.

Drawn structure perfectly reconstructs the given name.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B02— NAME→STRUCTURE

Level: Foundation | Given Name: 2-methylpropane

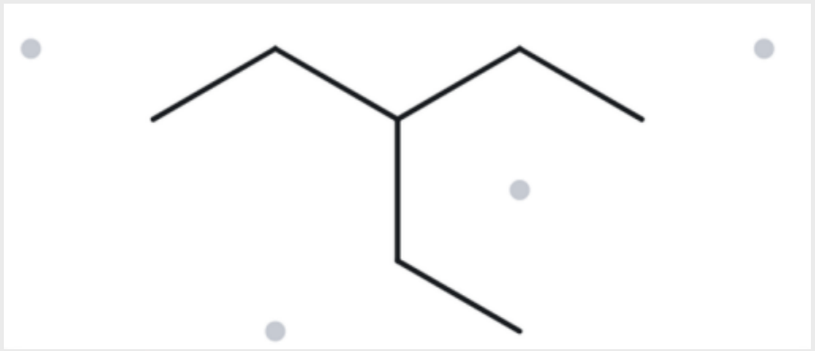
1. Name Decoded

Parent: pentane → 5-carbon chain.

Substituents: ethyl on C3.

Expected formula: 5 + 2 = 7 C; open chain → C7H16 .

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C7H16

Open chain rule: CnH2n+2 = C7H16 (Matches)

4. Back-Check

✓ Match

The Alkane returned: 3-ethylpentane

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B03— NAME→STRUCTURE

Level: Foundation | Given Name: 2,2-dimethylbutane

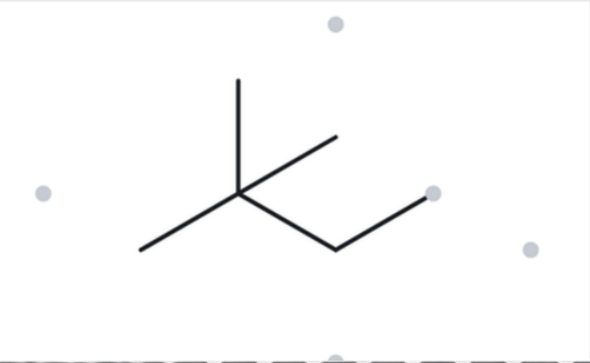
1. Name Decoded

Parent: butane → 4-carbon chain.

Substituents: two methyls on C2.

Expected formula: 4 + 2 = 6 C; open chain → C6H14 .

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C6H14

Open chain rule: CnH2n+2 = C6H14 (Matches)

4. Back-Check

✓ **Match**

The Alkane returned: 2,2-dimethylbutane.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B04— NAME→STRUCTURE

Level: Developing | Given Name: 3-ethyl-2-methylpentane

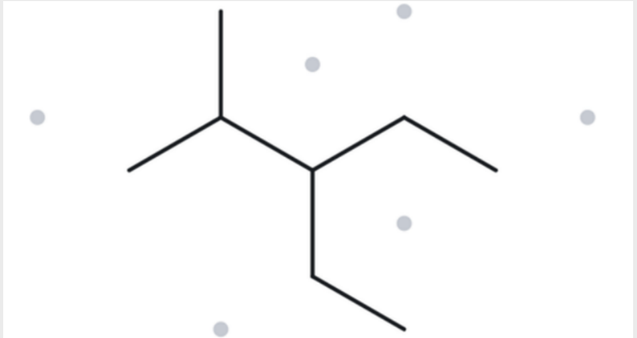
1. Name Decoded

Parent: pentane → 5-carbon chain.

Substituents: ethyl on C3, methyl on C2.

Expected formula: 5 + 2 + 1 = 8 C; open chain → C_8H_{18}.

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C_8H_{18}

Open chain rule: C_nH_{2n+2} = C_8H_{18} (Matches)

4. Back-Check

✓ **Match**

The Alkane returned: 3-ethyl-2-methylpentane.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B05— NAME→STRUCTURE

Level: Developing | Given Name: 1-chloro-2-methylpropane

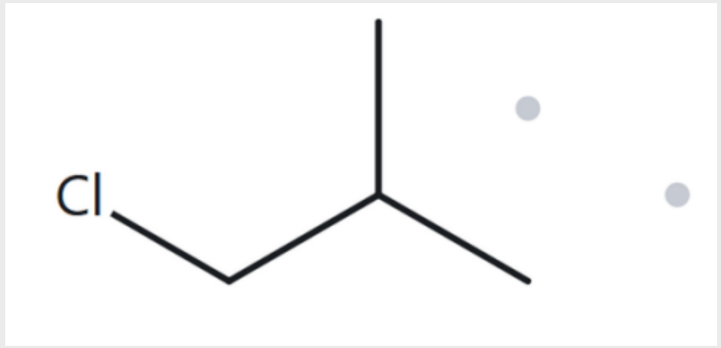
1. Name Decoded

Parent: propane → 3-carbon chain.

Substituents: chloro on C1, methyl on C2.

Expected formula: 3 + 1 = 4 C. Open chain, minus 1 H for Cl → C4H9Cl.

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C4H9Cl

Open chain rule: CnH2n+2 = C4H10 , swap 1 H for Cl = C4H9Cl (Matches)

4. Back-Check

✓ Match

The Alkane returned: 1-chloro-2-methylpropane.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B06— NAME→STRUCTURE

Level: Developing | Given Name: 1,2-dimethylcyclopentane

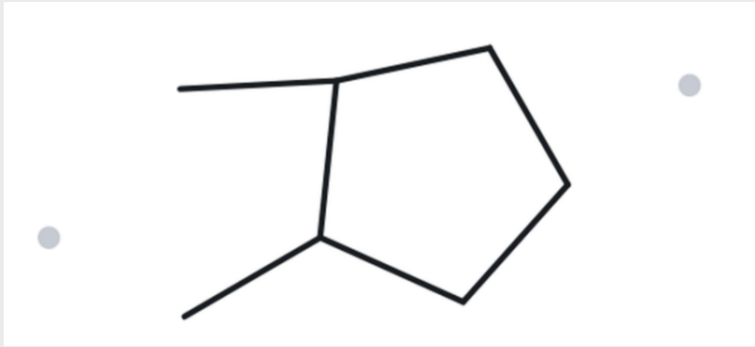
1. Name Decoded

Parent: cyclopentane → 5-carbon ring.

Substituents: methyl on C1, methyl on C2.

Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14} .

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C_7H_{14}

One ring rule: C_nH_{2n} = C_7H_{14} (Matches)

4. Back-Check

✓ **Match**

The Alkane returned: 1,2-dimethylcyclopentane.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B07— NAME→STRUCTURE

Level: Developing | Given Name: 4-ethyl-3,3-dimethylheptane

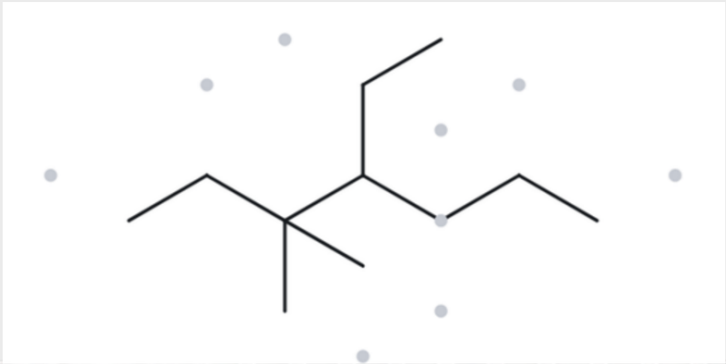
1. Name Decoded

Parent: heptane → 7-carbon chain.

Substituents: ethyl on C4, two methyls on C3.

Expected formula: 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$.

2. Bond-Line Structure



3. Formula Check

Counted from drawing: $C_{11}H_{24}$

Open chain rule: C_nH_{2n+2} = $C_{11}H_{24}$ (Matches)

4. Back-Check

✓ **Match**

The Alkane returned: 4-ethyl-3,3-dimethylheptane.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

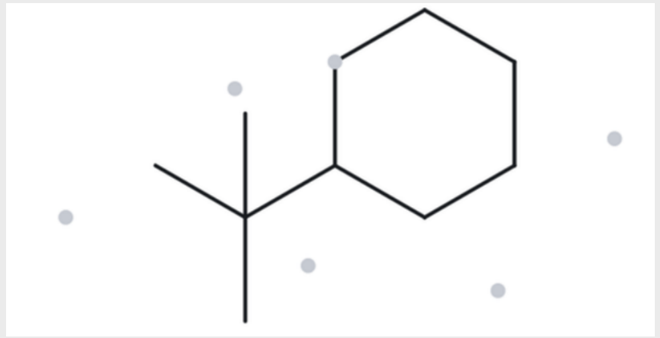
CARD B08— NAME→STRUCTURE

Level: Challenge | Given Name: tert-butylcyclohexane

1. Name Decoded

Parent: cyclohexane → 6-carbon ring.
Substituents: tert-butyl (4-carbon branched group) on C1.
Expected formula: 6 + 4 = 10 C; one ring → C10H20 .

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C10H20
One ring rule: CnH2n = C10H20 (Matches)

4. Back-Check

⚠ **Systematic Equivalent**
The Alkane returned: (1,1-dimethylethyl)cyclohexane. Both are completely correct; the tool returns the strict systematic name for the tert-butyl group.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B09— NAME→STRUCTURE

Level: Challenge | Given Name: 3-ethyl-1,1-dimethylcyclohexane

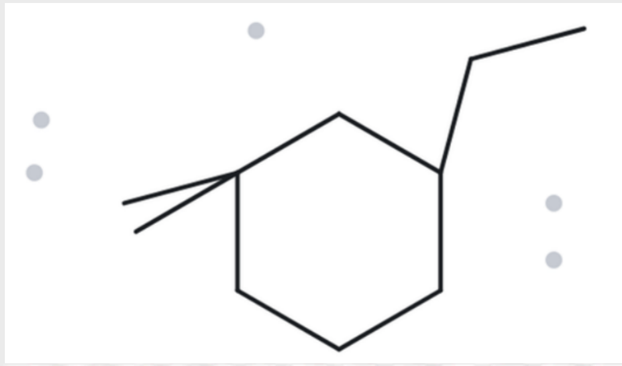
1. Name Decoded

Parent: cyclohexane → 6-carbon ring.

Substituents: two methyls on C1, ethyl on C3.

Expected formula: 6 + 2 + 2 = 10 C; one ring → C10H20 .

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C10H20

One ring rule: CnH2n = C10H20 (Matches)

4. Back-Check

✓**Match**

The Alkane returned: 3-ethyl-1,1-dimethylcyclohexane.

Priority and numbering correctly align.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded

CARD B10— NAME→STRUCTURE

Level: Challenge | Given Name: 2-ethylbutane

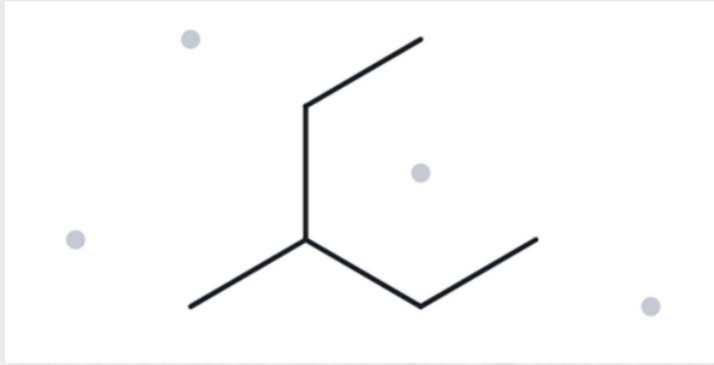
1. Name Decoded

Parent: butane → 4-carbon chain.

Substituents: ethyl on C2.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2. Bond-Line Structure



3. Formula Check

Counted from drawing: C_6H_{14}

Open chain rule: C_nH_{2n+2} = C_6H_{14} (Matches)

4. Back-Check

⚠ **Deliberate Mismatch (Error Found)**

The Alkane returned: 3-methylpentane. The given name violates the longest-chain rule. Appending an ethyl group to C2 of a 4-carbon chain creates a continuous 5-carbon chain (pentane), leaving a methyl group on C3.

Verification Strip

✅ parent length ✅ substituent positions ✅ formula agrees ✅ back-check recorded